

Supplementary Information for MolTrustBench

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Overview

This supplementary information summarizes the machine-readable source tables, quality-control summaries, and trust-card schema files used by MolTrustBench. The companion evidence archive is deposited at Zenodo: 10.5281/zenodo.20446372. Raw ChEMBL database dumps, local secrets, transient caches, and third-party model checkpoints are not redistributed; ChEMBL releases, public benchmark datasets, and public model checkpoints should be accessed through their original sources.

Source Tables and Artifacts

Table	Description	Claim	Rows	Supplement file
Table_S1	ChEMBL release-index summary	C1/C2	5	tables/Table_S1_chembl_release_counts.csv
Table_S2	Benchmark exposure summary across audited tasks	C1/C2	4	tables/Table_S2_benchmark_coverage_exposure_summary.csv
Table_S3	Core BBBP/ClinTox exposure-adjusted baseline metrics	C3	16	tables/Table_S3_slice_metrics.csv
Table_S4	BACE/Tox21 C3 expansion metrics	C3	30	tables/Table_S4_bace_tox21_c3_summary.csv
Table_S5	BACE/Tox21 sequence-family metrics	C3/C5	60	tables/Table_S5_bace_tox21_sequence_summary.csv
Table_S6	Exposure-adjusted CI summary	C3	80	tables/Table_S6_exposure_delta_ci.csv
Table_S7	Sequence-family CI summary	C3	24	tables/Table_S7_sequence_delta_ci.csv
Table_S8	ESOL/Lipophilicity regression CI summary	C3	24	tables/Table_S8_regression_delta_ci.csv
Table_S9	Endpoint-native ChEMBL temporal metrics	C3/C4	108	tables/Table_S9_chembl_endpoint_temporal_summary.csv
Table_S10	Endpoint SMILES-Transformer add-on metrics	C3	42	tables/Table_S10_chembl_endpoint_transformer_summary.csv
Table_S11	Train-label-shuffle null-control comparisons	C3	22	tables/Table_S11_label_shuffle_null_control_comparison.csv
Table_S12	CYP label-shuffle rescue null summary	C3	3	tables/Table_S12_cyp_extension_label_shuffle_null_summary.csv
Table_S13	Frozen MolFormer embedding summary	C3	57	tables/Table_S13_fm_embedding_summary.csv
Table_S14	Frozen ChemBERTa-100M embedding summary	C3	30	tables/Table_S14_chemberta_100m_fm_summary.csv
Table_S15	Assay-provenance task summary with DILI/hepatotoxicity	C4	6	tables/Table_S15_assay_provenance_task_summary_with_dili.csv
Table_S16	CYP1A2/CYP2C19 assay-provenance summary	C4	2	tables/Table_S16_cyp_endpoint_extension_assay_provenance_summary.csv
Table_S17	DILI/hepatotoxicity split and provenance boundary	C4	6	tables/Table_S17_dili_hepatotoxicity_split_summary.csv
Table_S18	DILI/hepatotoxicity sequence summary	C3/C4	12	tables/Table_S18_dili_hepatotoxicity_sequence_summary.csv
Table_S19	Model pretraining registry	C5	12	tables/Table_S19_model_pretraining_registry.csv
Table_S20	Current control and reporting checklist	C5	10	tables/Table_S20_required_controls.csv
Table_S21	Trust-card example source data	C5	4	tables/Table_S21_trust_card_examples.csv
Table_S22	Workflow artifact map	C5	7	tables/Table_S22_workflow_artifact_map.csv
Table_S23	Expected limitation and critical-failure accounting	C3/C5	6	tables/Table_S23_expected_limitations_summary.csv
Table_S24	Slice-validity master table	C3	12626	tables/Table_S24_slice_validity_master.csv
Table_S25	Standardization audit summary	C1/C2	13	tables/Table_S25_standardization_audit.csv
Table_S26	Assay-provenance flag dictionary	C4	6	tables/Table_S26_assay_flag_dictionary.csv
Table_S27	Assay-provenance example rows	C4	22	tables/Table_S27_assay_provenance_examples.csv
Table_S28	Machine-readable trust-card schema	C5	12	tables/Table_S28_trust_card_schema.csv
Table_S29	Figure source-data map	C5	10	tables/Table_S29_figure_source_data_map.csv
Table_S30	Protocol parameters and reproducibility artifacts	C5	10	tables/Table_S30_protocol_parameters.csv

Interpretation Notes

Tables S1–S2 report observable public-exposure lower bounds, not model-specific training inclusion. Tables S3–S12 summarize exposure-adjusted evaluation, repeated-run intervals, density controls, and train-label-shuffle negative controls. Tables S13–S14 are frozen-embedding applicability checks and do not support broad foundation-model conclusions. Tables S15–S18 summarize assay-provenance and endpoint-native toxicology evidence. Tables S23–S25 record expected slice limitations, metric validity, class-balance flags, and standardization accounting. Tables S26–S29 define provenance flags, give compact record-derived examples, provide the trust-card schema, and map figures to source data. Table S30 lists protocol parameters and reproducibility artifacts for rerunning the audit modules.

Rows marked as expected limitations, not auditable in the compact package, or missing class counts are excluded from inferential claims. Exposure-removed slices that become too small or one-class are reported as benchmark-usability limitations rather than performance evidence.